Practice workbook · The discriminant: equal, real or no real roots

Conditions on the discriminant of a quadratic, either given directly or obtained by substituting a line into a curve or rearranging a curve as a quadratic in x.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-05

1. Question 1 · part whole · 4 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part whole only.

1 Find the set of values of k for which the equation

$$2kx^2 + 5kx + 5k - 3 = 0 \qquad \text{where } k \neq 0$$

has real roots.

(4)

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(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 5 marks

2023 NOV · Paper 02

Further practice · Attempt part whole only.

1 The equation $kx^2 + 8x + 3k = 0$ where k is a constant, has real unequal roots.

Find the set of values of k giving your answer in an exact simplified form.

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(Total for Question 1 is 5 marks)

2. Question 1 · part whole · 5 marks · Working and answer

Working and answer

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3. Question 7 · part a · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

7 (a) Find the set of values of k for which the equation $kx^2 - 4x + 2k = 7$ has real roots

(4)

Given that the roots of the equation $kx^2 - 4x + 2k = 7$ are α and β,

(b) form a quadratic equation with roots $\frac{\alpha + 1}{\alpha}$ and $\frac{\beta + 1}{\beta}$

Give each coefficient in terms of k.

(8)

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3. Question 7 · part a · 4 marks · Working and answer

Working and answer

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4. Question 1 · part whole · 6 marks

2021 JUN · Paper 01

Further practice · Attempt part whole only.

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1 The quadratic equation

$$3(k + 2)x^2 + (k + 5)x + k = 0$$

has real roots.

Find the set of possible values of k.

(6)

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(Total for Question 1 is 6 marks)

4. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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5. Question 1 · part whole · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

1 The quadratic equation

$$kx^2 - (2k + 6)x + 16 = 0$$

has equal roots.

Find the two possible values of k

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(Total for Question 1 is 4 marks)

5. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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6. Question 2 · part whole · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

2 $f(x) = 2x^2 + (p - 1)x - 2p$ where p is a constant.

Find the set of values of p for which the equation $f(x) = 0$ has two distinct real roots.

(5)

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6. Question 2 · part whole · 5 marks

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Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 5 marks)

6. Question 2 · part whole · 5 marks · Working and answer

Working and answer

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7. Question 1 · part whole · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

1 $f(x) = 2x^2 + (k+8)x + k$

Show that for all values of k, the equation $f(x) = 0$ has distinct real roots.

(4)

(Total for Question 1 is 4 marks)

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7. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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8. Question 9 · part b · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

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8. Question 9 · part b · 4 marks · Working and answer

Working and answer

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9. Question 9 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

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(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

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9. Question 9 · part a · 3 marks · Working and answer

Working and answer

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10. Question 5 · part a · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

5 Given that k is a non-zero constant

curve C has equation $kx^2 - xy + (k+1)x = 1$

straight line l has equation $y = \frac{k}{2}x + 1$

The point A is the only point that lies on both C and l.

(a) Find the value of k

(6)

(b) Hence, find the coordinates of A.

(2)

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10. Question 5 · part a · 6 marks · Working and answer

Working and answer

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11. Question 10 · part c · 9 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

10 The curve C has equation $y = \dfrac{ax - 5}{b - x}$ where a and b are integers and $x \neq b$

One intersection of C with the coordinate axes is at the point with coordinates $\left(\frac{5}{4}, 0\right)$

The asymptote parallel to the y-axis has equation $x = 3$

(a) Find the value of a and the value of b

(2)

(b) Sketch C, showing clearly the asymptotes with their equations and the coordinates of the points of intersection with the coordinate axes.

(5)

The straight line l with equation $4y - 7x = k$ has no points of intersection with C

(c) Show, using algebra, that the range of possible values of k can be written as

$$m < k < n$$

where m and n are integers to be found.

(9)

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11. Question 10 · part c · 9 marks

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Further practice · Attempt part c only.

Question 10 continued

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11. Question 10 · part c · 9 marks

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Question 10 continued

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11. Question 10 · part c · 9 marks

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Question 10 continued

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11. Question 10 · part c · 9 marks

2024 JUN · Paper 01

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Question 10 continued

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(Total for Question 10 is 16 marks)

11. Question 10 · part c · 9 marks · Working and answer

Working and answer

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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12. Question 11 · part d · 7 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 11 continued

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(Total for Question 11 is 14 marks)

12. Question 11 · part d · 7 marks · Working and answer

Working and answer

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